



World University of Bangladesh

Lab Report 2

COURSE TITLE: Object Oriented Programming Lab

COURSE CODE: CSE 805

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Q. 1. Write a C++/Java program that performs the basic I/O using the concepts of class and objects. [No static initialization will be allowed. Any input must be through the constructor functions and the output through the member functions.]

Ans:

```
1.cpp x
1  /*1. Write a C++/Java program that performs the basic I/O using the concepts of class and objects.
2  [No static initialization will be allowed. Any input must be through the constructor functions and
3  the output through the member functions.]*/*
4
5  #include <iostream>
6  using namespace std;
7
8  class Person {
9  private:
10     string name;
11     int age;
12
13 public:
14     Person(string n, int a) : name(n), age(a) {}
15
16     void displayInfo() {
17         cout << "Name: " << name << ", Age: " << age << endl;
18     }
19 };
20 int main() {
21     string name;
22     int age;
23
24     cout << "Enter name: ";
25     cin >> name;
26     cout << "Enter age: ";
27     cin >> age;
28
29     Person person(name, age);
30     person.displayInfo();
31
32     return 0;
33 }
34
```

```
"C:\Users\User\Desktop\OOP" x + v
Enter name: Munim
Enter age: 23
Name: Munim, Age: 23

Process returned 0 (0x0)   execution time : 14.884 s
Press any key to continue.
```

Q. 2. Write a C++/Java program that uses reference variables as arguments to swap the values of a pair of integers using the concepts of class and objects. [No static initialization will be allowed. Any input must be through the constructor functions and the output through the member functions.]

Ans:

```
#include <iostream>
using namespace std;

class Swapper {
private:
    int &a;
    int &b;

public:
    Swapper(int &x, int &y) : a(x), b(y) {}

    void swapValues() {
        int temp = a;
        a = b;
        b = temp;
    }

    void displayValues() {
        cout << "a: " << a << ", b: " << b << endl;
    }
};
```

```
int main() {
    int num1, num2;

    cout << "Enter the first number: ";
    cin >> num1;
    cout << "Enter the second number: ";
    cin >> num2;

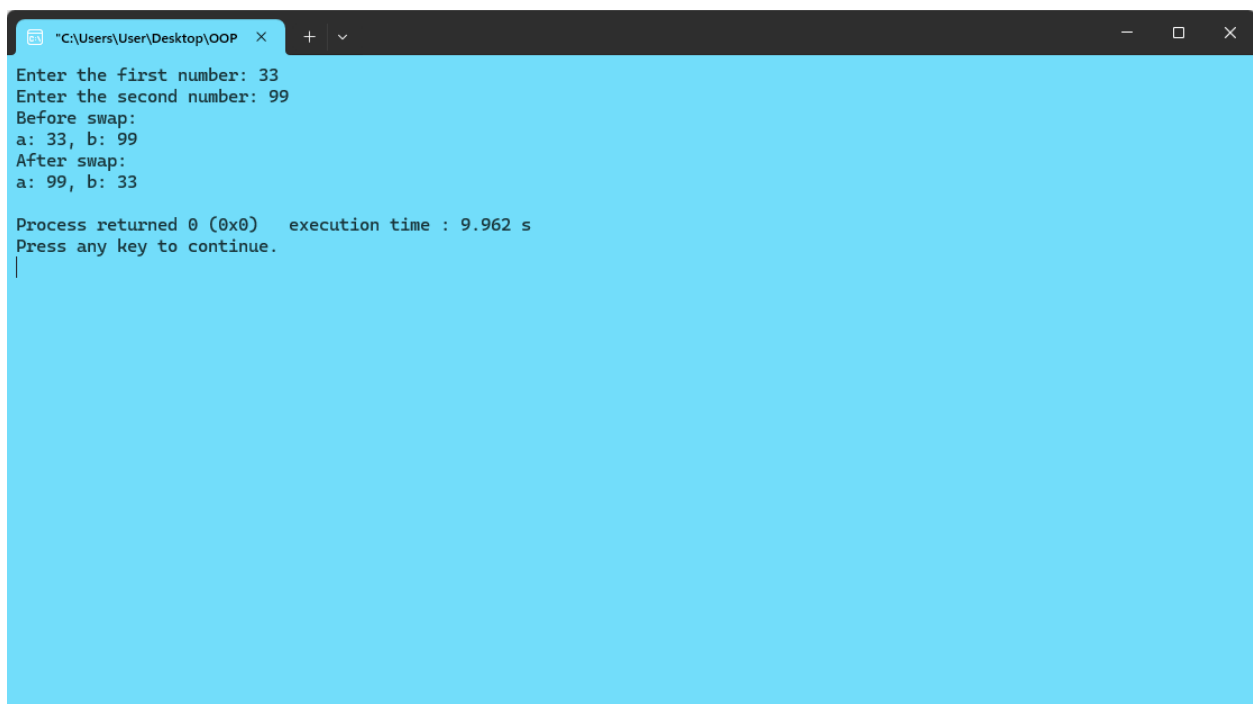
    Swapper swapper(num1, num2);

    cout << "Before swap:" << endl;
    swapper.displayValues();

    swapper.swapValues();

    cout << "After swap:" << endl;
    swapper.displayValues();

    return 0;
}
```



```
"C:\Users\User\Desktop\OOP" x + v
Enter the first number: 33
Enter the second number: 99
Before swap:
a: 33, b: 99
After swap:
a: 99, b: 33

Process returned 0 (0x0)   execution time : 9.962 s
Press any key to continue.
|
```

Q. 3. Write a C++/Java program that performs basic arithmetic operations using the concepts of class and objects. [No static initialization will be allowed. Any input must be through the constructor functions and the output through the member functions.]

Ans:

```
#include <iostream>
using namespace std;

class ArithmeticOperations {
private:
    double num1;
    double num2;
    double temp;

public:
    ArithmeticOperations(double x, double y) : num1(x), num2(y) {}

    double add() {
        return num1 + num2;
    }

    double subtract() {
        return num1 - num2;
    }
}
```

```

double multiply() {
    return num1 * num2;
}

double divide() {
    if (num2 != 0) {
        return num1 / num2;
    }
    else {
        cout << "Error: Division by zero!" << endl;
        return 0;
    }
}

void displayResults() {
    cout << "Addition: " << add() << endl;
    cout << "Subtraction: " << subtract() << endl;
    cout << "Multiplication: " << multiply() << endl;
    cout << "Division: " << divide() << endl;
}

```

```

};

int main() {
    double number1, number2;

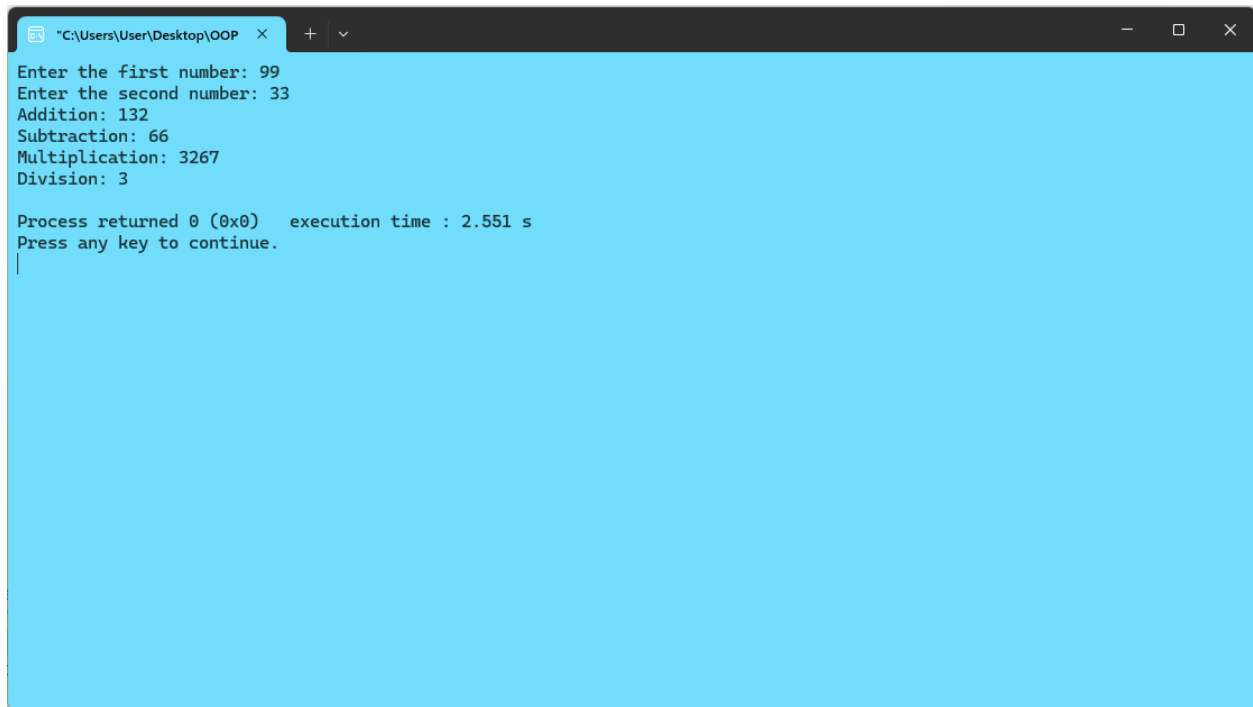
    cout << "Enter the first number: ";
    cin >> number1;
    cout << "Enter the second number: ";
    cin >> number2;

    ArithmeticOperations operations(number1, number2);

    operations.displayResults();

    return 0;
}

```



A screenshot of a Windows command prompt window with a light blue background. The window title bar shows the file path "C:\Users\User\Desktop\OOP" and standard window controls. The text inside the window shows the execution of a C++ program. It prompts for two numbers, 99 and 33, and then displays the results of addition, subtraction, multiplication, and division. The addition result is 132, subtraction is 66, multiplication is 3267, and division is 3. At the end, it shows the process returned 0 and the execution time was 2.551 seconds, followed by a prompt to press any key to continue.

```
"C:\Users\User\Desktop\OOP" x + v
Enter the first number: 99
Enter the second number: 33
Addition: 132
Subtraction: 66
Multiplication: 3267
Division: 3

Process returned 0 (0x0)   execution time : 2.551 s
Press any key to continue.
|
```

4. Write a C++/Java program that handles the students' records using the concepts of class and objects. [No static initialization will be allowed. Any input must be through the constructor functions and the output through the member functions.]

Ans:

```

#include <iostream>
#include <vector>
#include <string>
using namespace std;

class Student {
private:
    string name;
    int rollNumber;
    double marks;

public:
    Student(string n, int r, double m) : name(n), rollNumber(r), marks(m) {}

    void displayDetails() {
        cout << "Name: " << name << ", Roll Number: " << rollNumber << ", Marks: " << marks << endl;
    }

    string getName() {
        return name;
    }

    int getRollNumber() {
        return rollNumber;
    }

    double getMarks() {
        return marks;
    }
};

class StudentRecords {
private:
    vector<Student> students;

public:
    void addStudent(Student student) {
        students.push_back(student);
    }

    void displayAllStudents() {
        for (Student s : students) {
            s.displayDetails();
        }
    }
};

```

```

int main() {
    int n;
    cout << "Enter the number of students: ";
    cin >> n;

    StudentRecords records;

    for (int i = 0; i < n; ++i) {
        string name;
        int rollNumber;
        double marks;

        cout << "Enter details for student " << i + 1 << ":" << endl;
        cout << "Name: ";
        cin >> ws;
        getline(cin, name);
        cout << "Roll Number: ";
        cin >> rollNumber;
        cout << "Marks: ";
        cin >> marks;

        Student student(name, rollNumber, marks);
        records.addStudent(student);
    }

    cout << "\nAll Students' Records:" << endl;
    records.displayAllStudents();

    return 0;
}

```

```
"C:\Users\User\Desktop\OOP" x + v - □ ×
Enter the number of students: 3
Enter details for student 1:
Name: Munim
Roll Number: 3802
Marks: 12
Enter details for student 2:
Name: Imran
Roll Number: 3813
Marks: 14
Enter details for student 3:
Name: Akash
Roll Number: 3800
Marks: 13

All Students' Records:
Name: Munim, Roll Number: 3802, Marks: 12
Name: Imran, Roll Number: 3813, Marks: 14
Name: Akash, Roll Number: 3800, Marks: 13

Process returned 0 (0x0)   execution time : 155.773 s
Press any key to continue.
|
```

Q. 5. Write a C++/Java program that calculates the GPA for students using the concepts of class and objects. [No static initialization will be allowed. Any input must be through the constructor functions and the output through the member functions.]

Ans:

```
#include <iostream>
#include <vector>
#include <string>
using namespace std;

class Student {
private:
    string name;
    int rollNumber;
    vector<double> grades;
    vector<int> credits;

public:
    Student(string n, int r, vector<double> g, vector<int> c) : name(n), rollNumber(r), grades(g), credits(c) {}

    double calculateGPA() {
        double totalGradePoints = 0;
        int totalCredits = 0;

        for (size_t i = 0; i < grades.size(); ++i) {
            totalGradePoints += grades[i] * credits[i];
        }
    }
};
```

```

        totalCredits += credits[i];
    }

    if (totalCredits == 0) {
        cout << "Error: Total credits cannot be zero." << endl;
        return 0;
    }

    return totalGradePoints / totalCredits;
}

void displayDetails() {
    cout << "Name: " << name << ", Roll Number: " << rollNumber << ", GPA: " << calculateGPA() << endl;
}

};

int main() {
    int numberOfStudents;
    cout << "Enter the number of students: ";
    cin >> numberOfStudents;

    vector<Student> students;

    for (int i = 0; i < numberOfStudents; ++i) {
        string name;
        int rollNumber;
        int numberOfSubjects;

        cout << "Enter details for student " << i + 1 << ":" << endl;
        cout << "Name: ";
        cin >> ws;
        getline(cin, name);
        cout << "Roll Number: ";
        cin >> rollNumber;
        cout << "Enter the number of subjects: ";
        cin >> numberOfSubjects;

        vector<double> grades(numberOfSubjects);
        vector<int> credits(numberOfSubjects);

        for (int j = 0; j < numberOfSubjects; ++j) {
            cout << "Enter grade for subject " << j + 1 << ": ";

```

```

        cin >> grades[j];
        cout << "Enter credit for subject " << j + 1 << ": ";
        cin >> credits[j];
    }

    students.emplace_back(name, rollNumber, grades, credits);
}

cout << "\nAll Students' Records:" << endl;
for (Student& student : students) {
    student.displayDetails();
}

return 0;
}

```

```

C:\Users\User\Desktop\OOP > .\program.exe
Enter the number of students: 3
Enter details for student 1:
Name: Tawseef
Roll Number: 3
Enter the number of subjects: 2
Enter grade for subject 1: 4
Enter credit for subject 1: 3
Enter grade for subject 2: 4
Enter credit for subject 2: 1
Enter details for student 2:
Name: Nibir
Roll Number: 6
Enter the number of subjects: 2
Enter grade for subject 1: 4
Enter credit for subject 1: 3
Enter grade for subject 2: 4
Enter credit for subject 2: 1
Enter details for student 3:
Name: Munim
Roll Number: 9
Enter the number of subjects: 2
Enter grade for subject 1: 4
Enter credit for subject 1: 3
Enter grade for subject 2: 4
Enter credit for subject 2: 1

All Students' Records:
Name: Tawseef, Roll Number: 3, GPA: 4
Name: Nibir, Roll Number: 6, GPA: 4
Name: Munim, Roll Number: 9, GPA: 4

Process returned 0 (0x0)   execution time : 113.323 s
Press any key to continue.

```

Q. 6. Write a C++/Java program that calculates the CGPA for students using the concepts of class and objects. [No static initialization will be allowed. Any input must be through the constructor functions and the output through the member functions.]

Ans:

```
#include <iostream>
#include <vector>
#include <string>
using namespace std;

class Student {
private:
    string name;
    int rollNumber;
    vector<double> grades;
    vector<int> credits;

public:
    Student(string n, int r, vector<double> g, vector<int> c) : name(n), rollNumber(r), grades(g), credits(c) {}
    double calculateCGPA() {
        double totalGradePoints = 0;
        int totalCredits = 0;

        for (size_t i = 0; i < grades.size(); ++i) {
            totalGradePoints += grades[i] * credits[i];
            totalCredits += credits[i];
        }

        if (totalCredits == 0) {
            cout << "Error: Total credits cannot be zero." << endl;
            return 0;
        }

        return totalGradePoints / totalCredits;
    }

    void displayDetails() {
        cout << "Name: " << name << ", Roll Number: " << rollNumber << ", CGPA: " << calculateCGPA() << endl;
    }
};

class School {
private:
    vector<Student> students;

public:
    void addStudent(Student student) {
        students.push_back(student);
    }

    void displayAllStudents() {
```

```

    for (Student& s : students) {
        s.displayDetails();
    }
};

int main() {
    int numberOfStudents;

    cout << "Enter the number of students: ";
    cin >> numberOfStudents;

    School school;

    for (int i = 0; i < numberOfStudents; ++i) {
        string name;
        int rollNumber;
        int numberOfSubjects;

        cout << "Enter details for student " << i + 1 << ":" << endl;
        cout << "Name: ";
        cin >> ws;
        getline(cin, name);
        cout << "Roll Number: ";

        cin >> rollNumber;
        cout << "Enter the number of subjects: ";
        cin >> numberOfSubjects;

        vector<double> grades(numberOfSubjects);
        vector<int> credits(numberOfSubjects);

        for (int j = 0; j < numberOfSubjects; ++j) {
            cout << "Enter grade for subject " << j + 1 << ": ";
            cin >> grades[j];
            cout << "Enter credit for subject " << j + 1 << ": ";
            cin >> credits[j];
        }

        Student student(name, rollNumber, grades, credits);
        school.addStudent(student);
    }

    cout << "\nAll Students' Records:" << endl;
    school.displayAllStudents();

    return 0;
}

```

```
"C:\Users\User\Desktop\OOP" x + v
Enter the number of students: 2
Enter details for student 1:
Name: Munim
Roll Number: 3
Enter the number of subjects: 2
Enter grade for subject 1: 4
Enter credit for subject 1: 3
Enter grade for subject 2: 4
Enter credit for subject 2: 1
Enter details for student 2:
Name: Riddi
Roll Number: 4
Enter the number of subjects: 2
Enter grade for subject 1: 4
Enter credit for subject 1: 3
Enter grade for subject 2: 4
Enter credit for subject 2: 1

All Students' Records:
Name: Munim, Roll Number: 3, CGPA: 4
Name: Riddi, Roll Number: 4, CGPA: 4

Process returned 0 (0x0)   execution time : 79.115 s
Press any key to continue.
|
```

Q. 7. Write a C++/Java program that checks whether a given integer prime or not using the concepts of class and objects. [No static initialization will be allowed. Any input must be through the constructor functions and the output through the member functions.]

Ans:

```

#include <iostream>
using namespace std;

class PrimeChecker {
private:
    int number;

public:
    PrimeChecker(int num) : number(num) {}

    bool isPrime() {
        if (number <= 1) return false;
        if (number <= 3) return true;
        if (number % 2 == 0 || number % 3 == 0) return false;

        for (int i = 5; i * i <= number; i += 6) {
            if (number % i == 0 || number % (i + 2) == 0) return false;
        }
        return true;
    }

    void displayResult() {
        if (isPrime()) {
            cout << number << " is a prime number." << endl;

            } else {
                cout << number << " is not a prime number." << endl;
            }
        }
    };

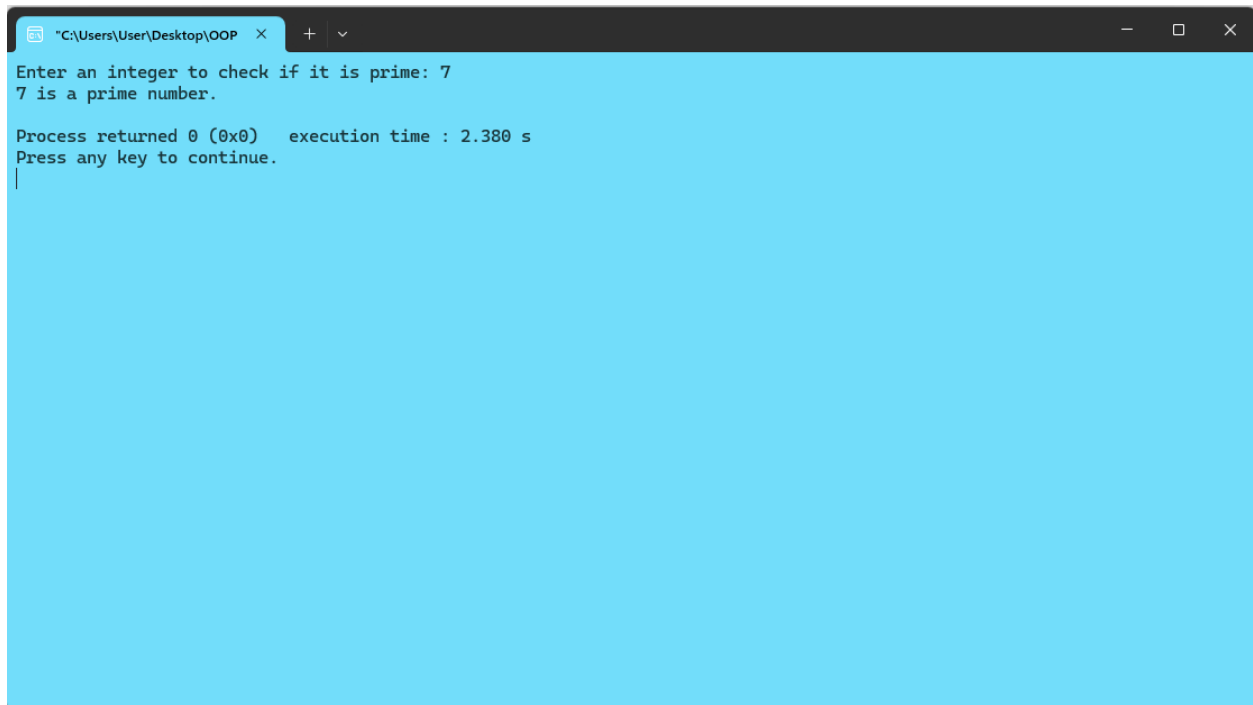
int main() {
    int number;

    cout << "Enter an integer to check if it is prime: ";
    cin >> number;

    PrimeChecker checker(number);
    checker.displayResult();

    return 0;
}

```



```
"C:\Users\User\Desktop\OOP" x + - x
Enter an integer to check if it is prime: 7
7 is a prime number.

Process returned 0 (0x0)   execution time : 2.380 s
Press any key to continue.
|
```

Q. 8. Write a C++/Java program that finds the factorial of a given number using the concepts of class and objects. [No static initialization will be allowed. Any input must be through the constructor functions and the output through the member functions.]

Ans:

```
#include <iostream>
using namespace std;

class FactorialCalculator {
private:
    int number;

public:
    FactorialCalculator(int num) : number(num) {}

    long long calculateFactorial() {
        if (number < 0) {
            cout << "Error: Factorial of a negative number is not defined." << endl;
            return -1;
        }
        long long factorial = 1;
        for (int i = 1; i <= number; ++i) {
            factorial *= i;
        }
        return factorial;
    }

    void displayResult() {
        long long result = calculateFactorial();

        if (result != -1) {
            cout << "The factorial of " << number << " is " << result << "." << endl;
        }
    }
};

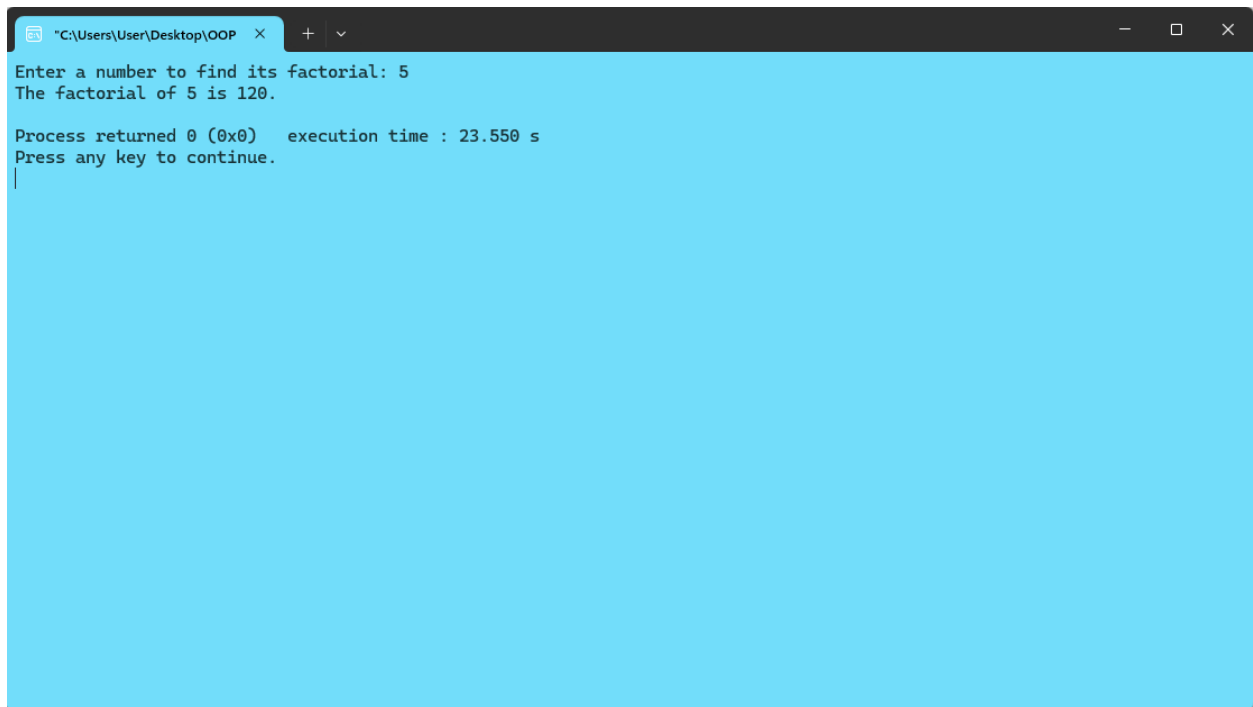
int main() {
    int number;

    cout << "Enter a number to find its factorial: ";
    cin >> number;

    FactorialCalculator calculator(number);

    calculator.displayResult();

    return 0;
}
```

A screenshot of a Windows command prompt window with a light blue background. The window title bar shows the file path "C:\Users\User\Desktop\OOP" and standard window controls. The text inside the window shows a program that prompts for a number, calculates its factorial, and displays the result. The input is 5, and the output is 120. It also shows the execution time and a prompt to press a key to continue.

```
"C:\Users\User\Desktop\OOP" x + v
Enter a number to find its factorial: 5
The factorial of 5 is 120.

Process returned 0 (0x0)   execution time : 23.550 s
Press any key to continue.
|
```

Q. 9. Write a C++/Java program that calculates the summation of any given series of numbers using the concepts of class and objects. [No static initialization will be allowed. Any input must be through the constructor functions and the output through the member functions.]

Ans:

```

#include <iostream>
#include <vector>
using namespace std;

class SeriesSummation {
private:
    vector<int> numbers;

public:
    SeriesSummation(const vector<int>& nums) : numbers(nums) {}

    int calculateSum() {
        int sum = 0;
        for (int num : numbers) {
            sum += num;
        }
        return sum;
    }

    void displayResult() {
        int result = calculateSum();
        cout << "The summation of the series is " << result << "." << endl;
    }
};

int main() {
    int n;
    cout << "Enter the number of elements in the series: ";
    cin >> n;

    vector<int> numbers(n);

    cout << "Enter the elements of the series:" << endl;
    for (int i = 0; i < n; ++i) {
        cin >> numbers[i];
    }

    SeriesSummation series(numbers);
    series.displayResult();

    return 0;
}

```

```
"C:\Users\User\Desktop\OOP" x + v
Enter the number of elements in the series: 9
Enter the elements of the series:
5
7
8
9
4
5
6
4
5
The summation of the series is 53.

Process returned 0 (0x0)   execution time : 15.053 s
Press any key to continue.
|
```

Q. 10. Write a C++/Java program that uses static data members using the concepts of class and objects. [No static initialization will be allowed. Any input must be through the constructor functions and the output through the member functions.]

Ans:

```

#include <iostream>
using namespace std;

class ObjectCounter {
private:
    static int objectCount;

public:
    ObjectCounter() {
        ++objectCount;
    }

    ~ObjectCounter() {
        --objectCount;
    }

    static int getObjectCount() {
        return objectCount;
    }

    void displayObjectCount() {
        cout << "Current number of objects: " << getObjectCount() << endl;
    }
};

int ObjectCounter::objectCount = 0;

int main() {
    cout << "Creating first object..." << endl;
    ObjectCounter obj1;
    obj1.displayObjectCount();

    cout << "Creating second object..." << endl;
    ObjectCounter obj2;
    obj2.displayObjectCount();

    cout << "Creating third object..." << endl;
    ObjectCounter obj3;
    obj3.displayObjectCount();

    cout << "Destroying second object..." << endl;
    {
        ObjectCounter obj4;
        obj4.displayObjectCount();
    }
    cout << "Second object destroyed." << endl;
    obj1.displayObjectCount();
    obj2.displayObjectCount();
    obj3.displayObjectCount();

    return 0;
}

```

```
"C:\Users\User\Desktop\OOP" x + v
Creating first object...
Current number of objects: 1
Creating second object...
Current number of objects: 2
Creating third object...
Current number of objects: 3
Destroying second object...
Current number of objects: 4
Second object destroyed.
Current number of objects: 3
Current number of objects: 3
Current number of objects: 3

Process returned 0 (0x0)   execution time : 0.392 s
Press any key to continue.
|
```

Q. 11. Write a C++/Java program that uses static member functions using the concepts of class and objects. [No static initialization will be allowed. Any input must be through the constructor functions and the output through the member functions.]

Ans:

```

#include <iostream>
using namespace std;

class StaticDemo {
private:
    static int instanceCount;

public:
    StaticDemo() {
        ++instanceCount;
    }

    ~StaticDemo() {
        --instanceCount;
    }

    static int getInstanceCount() {
        return instanceCount;
    }

    static void displayInstanceCount() {
        cout << "Number of instances: " << getInstanceCount() << endl;
    }
};

int StaticDemo::instanceCount = 0;

int main() {
    cout << "Creating first instance..." << endl;
    StaticDemo obj1;
    StaticDemo::displayInstanceCount();

    cout << "Creating second instance..." << endl;
    StaticDemo obj2;
    StaticDemo::displayInstanceCount();

    cout << "Creating third instance..." << endl;
    StaticDemo obj3;
    StaticDemo::displayInstanceCount();

    cout << "Destroying second instance..." << endl;
    {
        StaticDemo obj4;
        StaticDemo::displayInstanceCount();
    }
    cout << "Second instance destroyed." << endl;
    StaticDemo::displayInstanceCount();

    return 0;
}

```

```
"C:\Users\User\Desktop\OOP" x + v
Creating first instance...
Number of instances: 1
Creating second instance...
Number of instances: 2
Creating third instance...
Number of instances: 3
Destroying second instance...
Number of instances: 4
Second instance destroyed.
Number of instances: 3

Process returned 0 (0x0)   execution time : 0.416 s
Press any key to continue.
|
```

Q. 12. Write a C++/Java program that implements the concept of constructor overloading.

Ans:

```
#include <iostream>
using namespace std;
class Box {
private:
    double length;
    double width;
    double height;
public:
    Box() : length(1.0), width(1.0), height(1.0) {
        cout << "Default constructor called." << endl;
    }

    Box(double side) : length(side), width(side), height(side) {
        cout << "Parameterized constructor with one argument called." << endl;
    }

    Box(double l, double w, double h) : length(l), width(w), height(h) {
        cout << "Parameterized constructor with three arguments called." << endl;
    }

    void displayDimensions() {
        cout << "Length: " << length << ", Width: " << width << ", Height: " << height << endl;
    }
};
```

```

int main() {

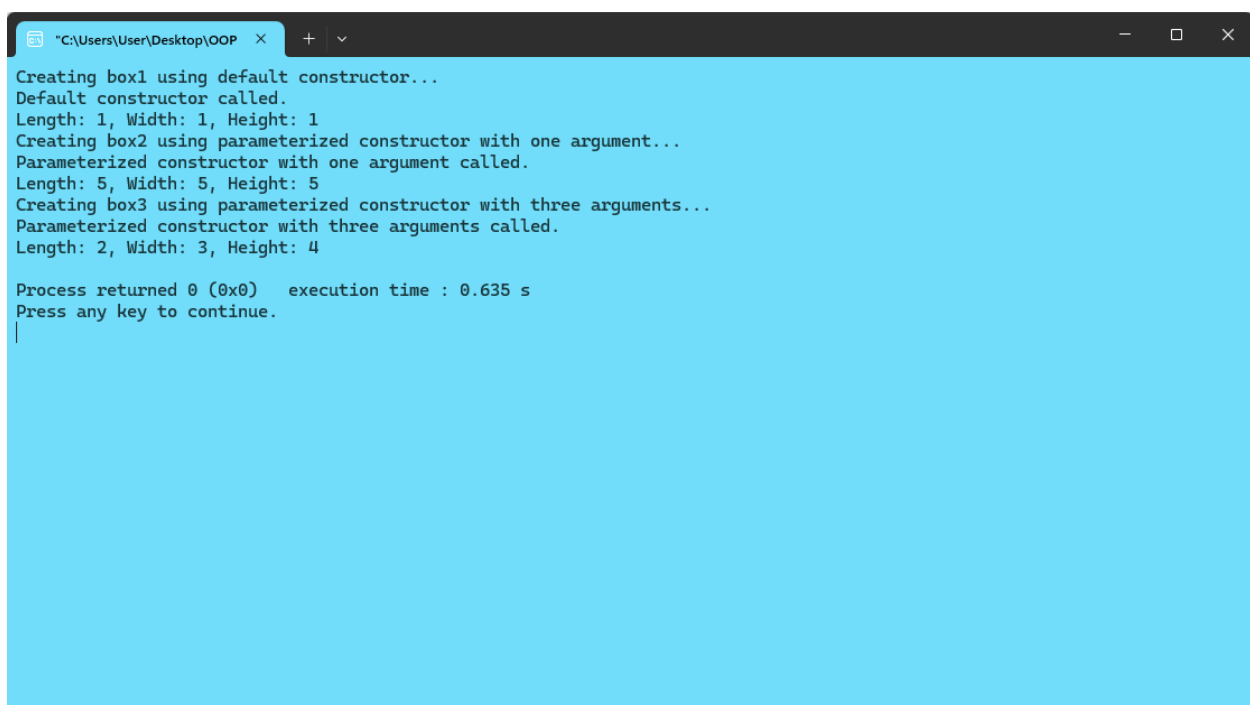
    cout << "Creating box1 using default constructor..." << endl;
    Box box1;
    box1.displayDimensions();

    cout << "Creating box2 using parameterized constructor with one argument..." << endl;
    Box box2(5.0);
    box2.displayDimensions();

    cout << "Creating box3 using parameterized constructor with three arguments..." << endl;
    Box box3(2.0, 3.0, 4.0);
    box3.displayDimensions();

    return 0;
}

```



The screenshot shows a Windows console window with the following output:

```

Creating box1 using default constructor...
Default constructor called.
Length: 1, Width: 1, Height: 1
Creating box2 using parameterized constructor with one argument...
Parameterized constructor with one argument called.
Length: 5, Width: 5, Height: 5
Creating box3 using parameterized constructor with three arguments...
Parameterized constructor with three arguments called.
Length: 2, Width: 3, Height: 4

Process returned 0 (0x0)   execution time : 0.635 s
Press any key to continue.
|

```

Q. 13. Write a C++/Java program that implements the concept of copy constructor.

Ans:

```
#include <iostream>
#include <cstring>

class MyString {
private:
    char* str;
public:
    MyString(const char* s = nullptr) {
        if (s) {
            str = new char[strlen(s) + 1];
            strcpy(str, s);
        } else {
            str = new char[1];
            str[0] = '\0';
        }
    }

    MyString(const MyString& source) {
        str = new char[strlen(source.str) + 1];
        strcpy(str, source.str);
    }

    ~MyString() {
        delete[] str;
    }

    void display() const {
        std::cout << str << std::endl;
    }

    const char* getStr() const {
        return str;
    }
};

int main() {
    char input[100];
    std::cout << "Enter a string: ";
    std::cin.getline(input, 100);

    MyString str1(input);
    MyString str2 = str1;

    std::cout << "Original string: ";
    str1.display();

    std::cout << "Copied string: ";
    str2.display();

    return 0;
}
```

```
"C:\Users\User\Desktop\OOP" x + v
Enter a string: I'm Munim
Original string: I'm Munim
Copied string: I'm Munim

Process returned 0 (0x0)   execution time : 22.547 s
Press any key to continue.
|
```

Q. 14. Write a C++/Java program that implements the concept run-time function overloading.

Ans:

```
#include <iostream>
#include <memory>

class Shape {
public:
    virtual void draw() const {
        std::cout << "Drawing a shape." << std::endl;
    }
    virtual ~Shape() = default;
};

class Circle : public Shape {
public:
    void draw() const override {
        std::cout << "Drawing a circle." << std::endl;
    }
};

class Rectangle : public Shape {
public:
    void draw() const override {
        std::cout << "Drawing a rectangle." << std::endl;
    }
};
```

```

void drawShape(const Shape& shape) {
    shape.draw();
}

int main() {
    std::unique_ptr<Shape> shape;
    int choice;

    std::cout << "Enter the shape to draw (1 for Circle, 2 for Rectangle): ";
    std::cin >> choice;

    if (choice == 1) {
        shape = std::make_unique<Circle>();
    } else if (choice == 2) {
        shape = std::make_unique<Rectangle>();
    } else {
        std::cout << "Invalid choice! Drawing default shape." << std::endl;
        shape = std::make_unique<Shape>();
    }

    drawShape(*shape);

    return 0;
}

```

```

C:\Users\User\Desktop\OOP >
Enter the shape to draw (1 for Circle, 2 for Rectangle): 1
Drawing a circle.

Process returned 0 (0x0)   execution time : 9.641 s
Press any key to continue.

```

Q. 15. Write a C++/Java program that implements the concept of constructor with default arguments.

Ans:

```
#include <iostream>
using namespace std;

class Rectangle {
private:
    int width;
    int height;

public:
    Rectangle(int w = 1, int h = 1) {
        width = w;
        height = h;
    }

    int area() {
        return width * height;
    }

    void display() {
        cout << "Width: " << width << ", Height: " << height << endl;
    }
};

int main() {
    Rectangle rect1;
    Rectangle rect2(5);
    Rectangle rect3(4, 7);

    cout << "Rectangle 1: ";
    rect1.display();
    cout << "Area: " << rect1.area() << endl;

    cout << "Rectangle 2: ";
    rect2.display();
    cout << "Area: " << rect2.area() << endl;

    cout << "Rectangle 3: ";
    rect3.display();
    cout << "Area: " << rect3.area() << endl;

    return 0;
}
```

```
"C:\Users\User\Desktop\OOP" x + - x
Rectangle 1: Width: 1, Height: 1
Area: 1
Rectangle 2: Width: 5, Height: 1
Area: 5
Rectangle 3: Width: 4, Height: 7
Area: 28

Process returned 0 (0x0)   execution time : 0.672 s
Press any key to continue.
|
```

Q. 16. Write a C++/Java program that implements the concept of destructor.

Ans:

```
#include <iostream>
using namespace std;

class Rectangle {
private:
    int width;
    int height;

public:
    Rectangle(int w, int h) {
        width = w;
        height = h;
        cout << "Rectangle created with width = " << width << " and height = " << height << endl;
    }

    ~Rectangle() {
        cout << "Rectangle destroyed with width = " << width << " and height = " << height << endl;
    }

    int area() {
        return width * height;
    }
}
```

```

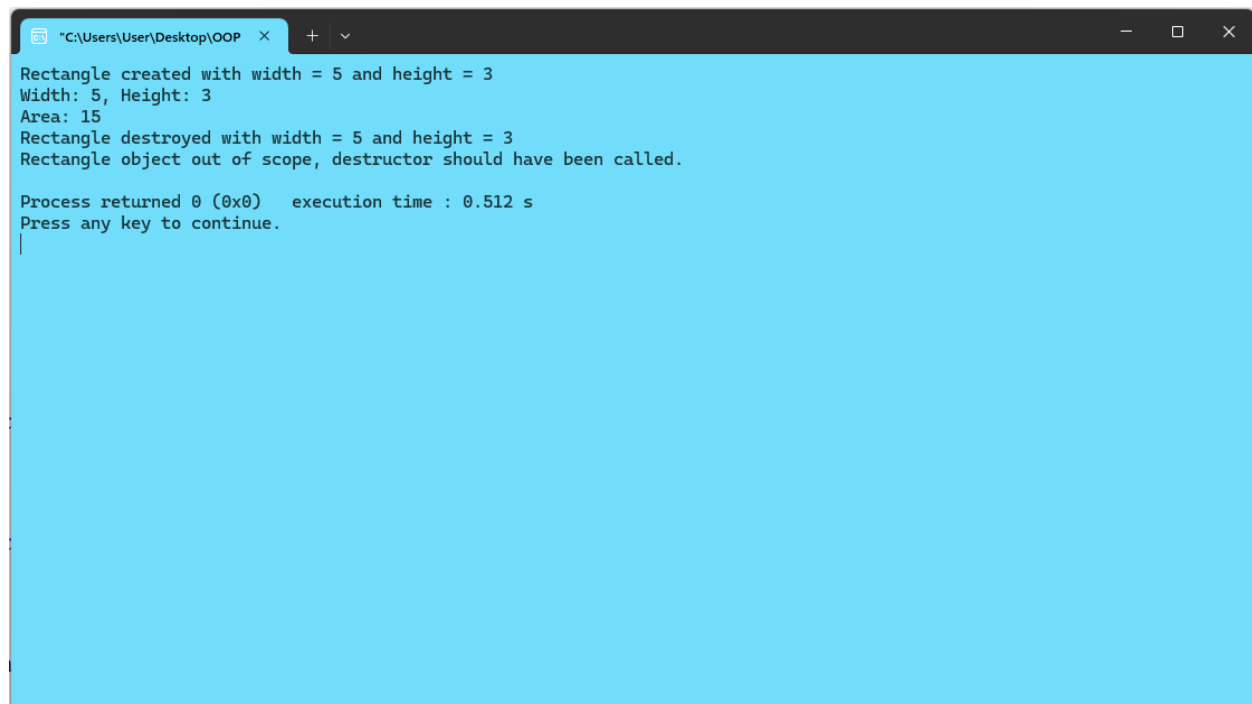
    void display() {
        cout << "Width: " << width << ", Height: " << height << endl;
    }
};

int main() {
    {
        Rectangle rect1(5, 3);
        rect1.display();
        cout << "Area: " << rect1.area() << endl;
    }

    cout << "Rectangle object out of scope, destructor should have been called." << endl;

    return 0;
}

```



```

"C:\Users\User\Desktop\OOP" x
+ v
Rectangle created with width = 5 and height = 3
Width: 5, Height: 3
Area: 15
Rectangle destroyed with width = 5 and height = 3
Rectangle object out of scope, destructor should have been called.

Process returned 0 (0x0)   execution time : 0.512 s
Press any key to continue.
|

```